Neural Networks and Function Approximation

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Overview

Outline

The Challenge: 1D Consolidation Settlement

Problem: Predict settlement from sparse field measurements.
Terzaghi consolidation theory:

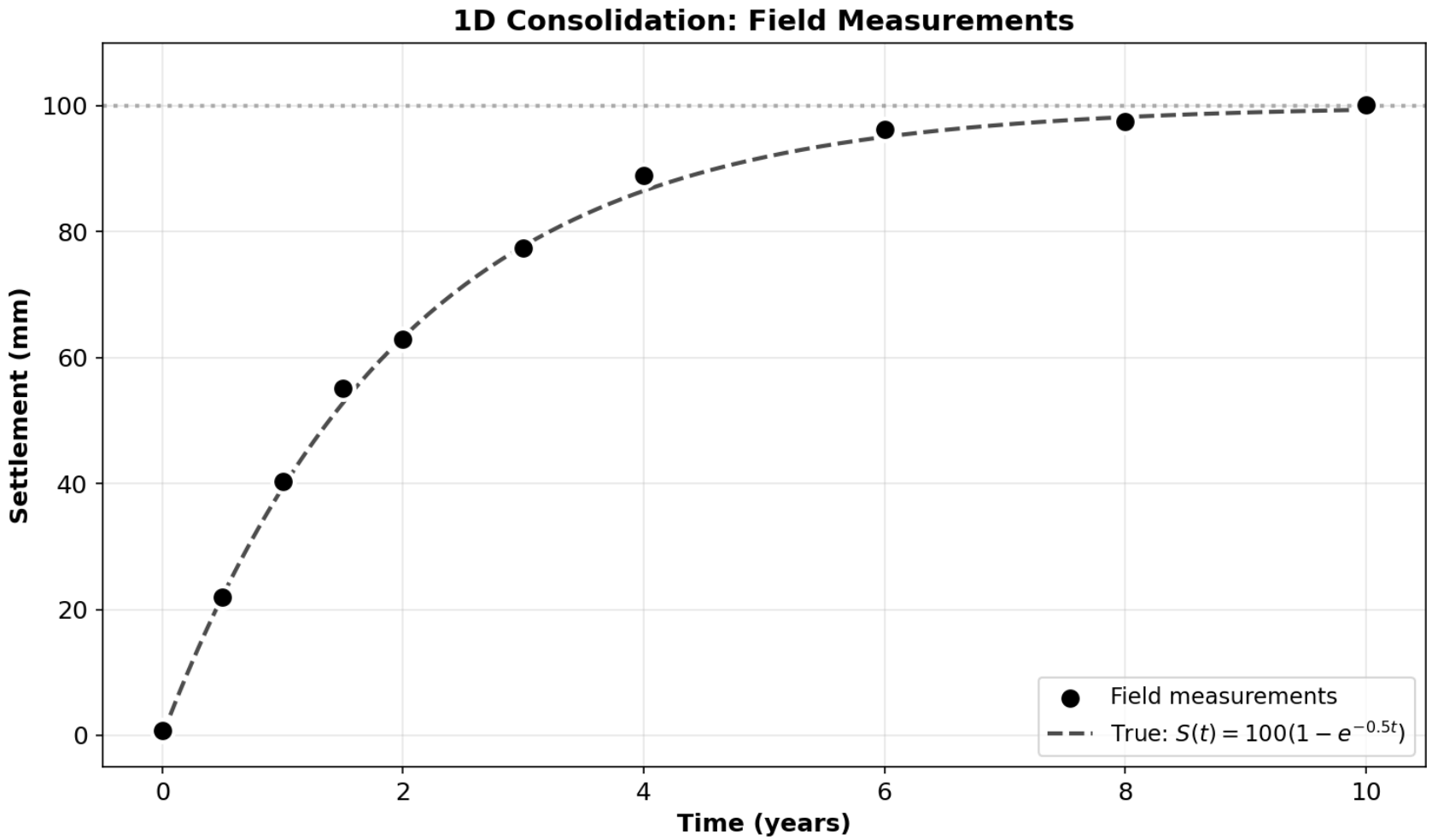
$$S(t) = S_{\text{final}}(1 - e^{-\alpha t})$$

Parameters:

- $S_{\text{final}} = 100$ mm (ultimate settlement)
- $\alpha = 0.5$ /year (consolidation coefficient)
- Sparse data: 10 measurements over 10 years

Goal: Neural network learns continuous $S(t)$ from discrete noisy measurements.

Consolidation: Field Measurements



Sparse field measurements with noise

Challenge: Nonlinear, bounded physical problem with limited data.

Outline

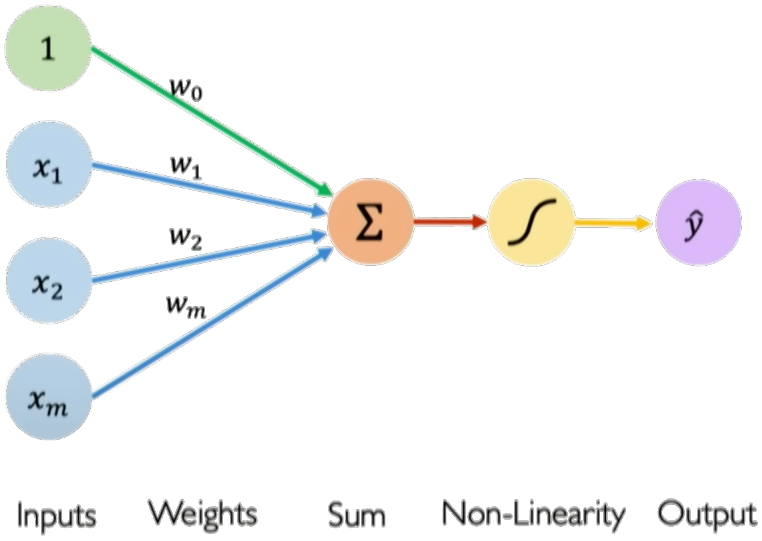
The Perceptron: Basic Building Block

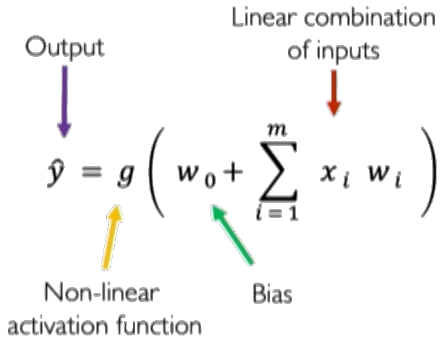
A perceptron computes:

$$z = \mathbf{w}^T \mathbf{x} + b = \sum_{i=1}^{n} w_i x_i + b$$

$$\hat{y} = g(z)$$

where g is the activation function.





Perceptron architecture

Key components: Input vector, weights, bias, activation function, output.

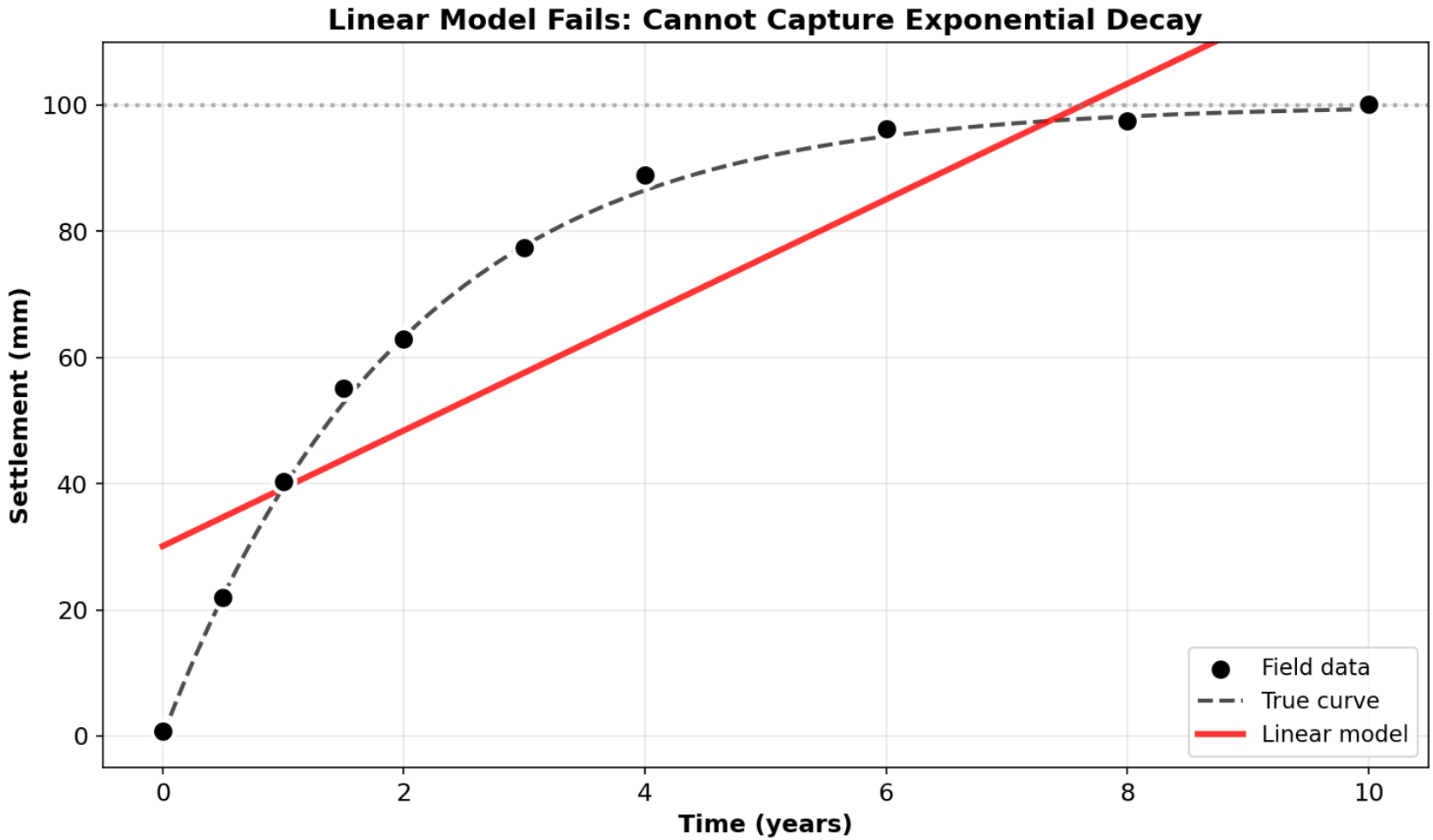
The Critical Role of Nonlinearity

Without activation functions: Multiple linear layers collapse to single linear transformation.
Consider two linear layers:

$$\begin{aligned} h_1 &= W_1 x + b_1 \\ h_2 &= W_2 h_1 + b_2 = W_2(W_1 x + b_1) + b_2 \\ &= (W_2 W_1)x + (W_2 b_1 + b_2) \end{aligned}$$

This is equivalent to: $h_2 = W_{eq} x + b_{eq}$ (still linear!)
Problem: Linear networks can only learn $y = mx + b$, but $\sin(\pi x)$ is curved!

Linear Model Fails on Consolidation



Linear model $\hat{S} = wt + b$ cannot capture exponential decay

Conclusion: Nonlinearity is essential for complex function approximation.

Common Activation Functions

Sigmoid: $\sigma(x) = \frac{1}{1+e^{-x}}$ (squashes to (0,1))

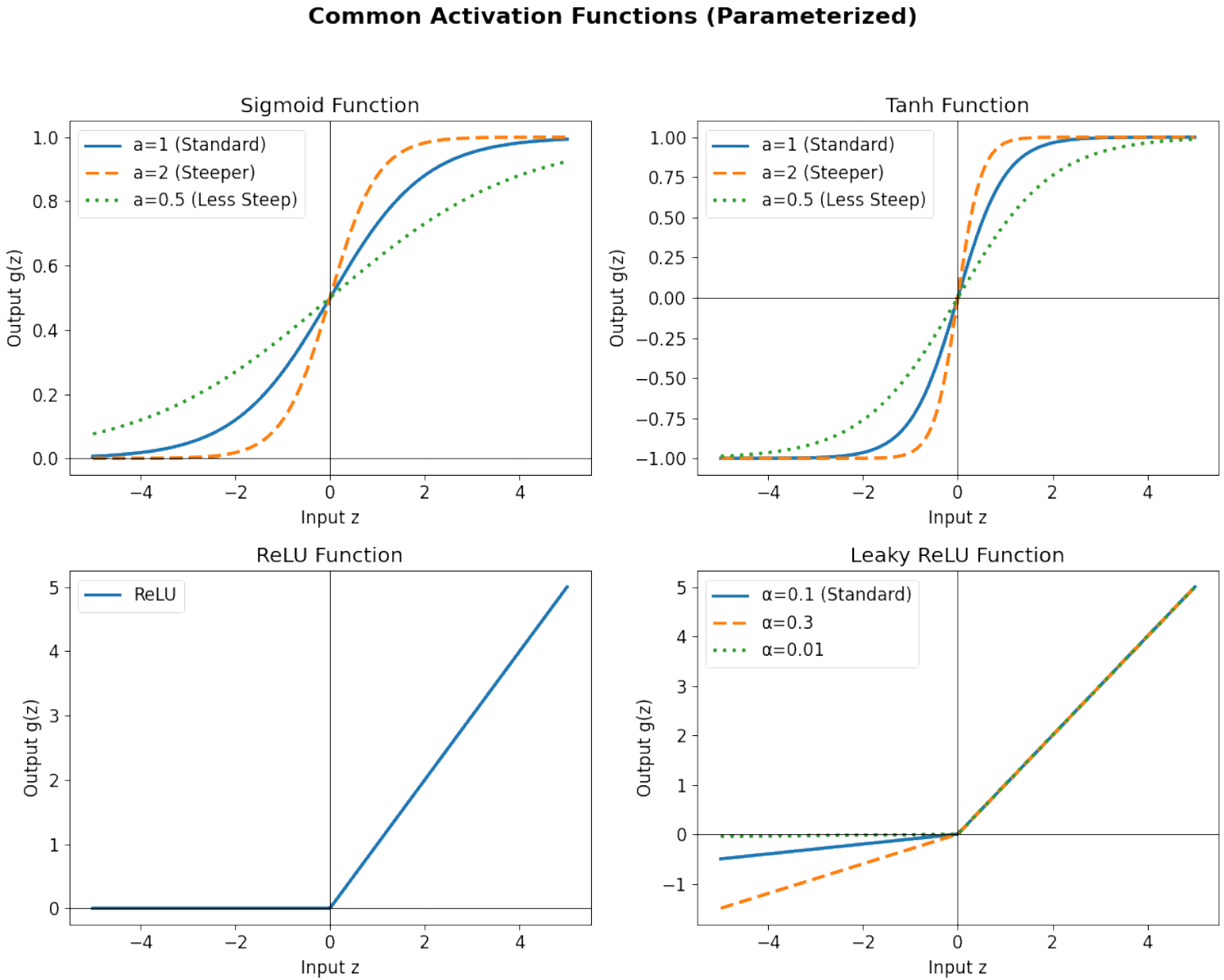
Tanh: $\tanh(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$ (squashes to (-1,1))

ReLU: $f(x) = \max(0, x)$ (efficient, prevents vanishing gradients)

Leaky ReLU: $f(x) = \max(\alpha x, x)$ (prevents dead neurons)

▸ ReLU Demo

Common Activation Functions



Outline

Universal Approximation Theorem

Theorem (Cybenko, 1989): A single hidden layer network with sufficient neurons can approximate any continuous function to arbitrary accuracy.

$$F(x) = \sum_{i=1}^{N} w_i \sigma(v_i x + b_i) + w_0$$

Mathematical statement: For any continuous $f : [0, 1] \to \mathbb{R}$ and $\epsilon > 0$, there exists N and parameters such that $|F(x) - f(x)| < \epsilon$ for all $x \in [0, 1]$.

Key Questions:

- How many neurons N do we need?
- Is this practical?
- Can we verify experimentally?

Single Hidden Layer Architecture

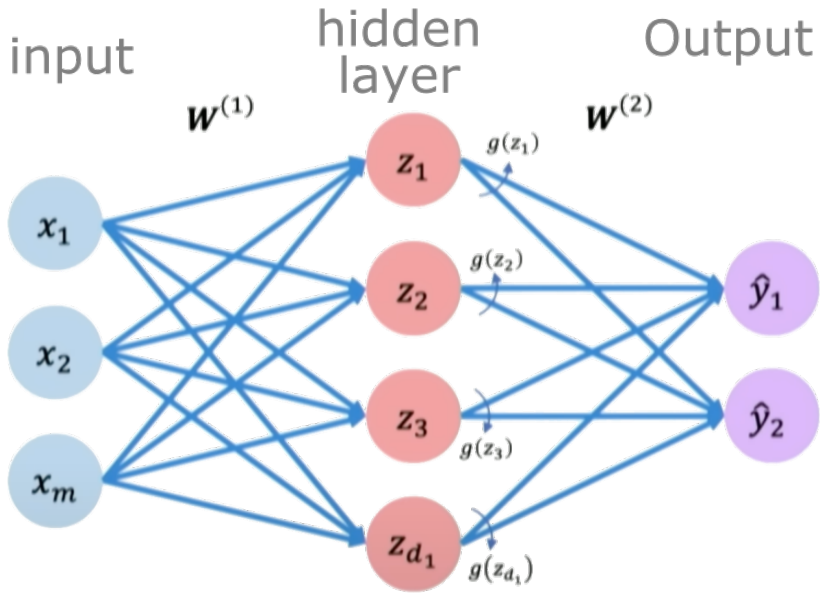
For 1D input x, a single-layer network with N_h hidden neurons:

$$\mathbf{z}^{(1)} = W^{(1)}x + \mathbf{b}^{(1)} \quad \text{(pre-activation)}$$

$$\mathbf{h} = g(\mathbf{z}^{(1)}) \quad \text{(hidden layer output)}$$

$$z^{(2)} = W^{(2)}\mathbf{h} + b^{(2)} \quad \text{(output layer)}$$

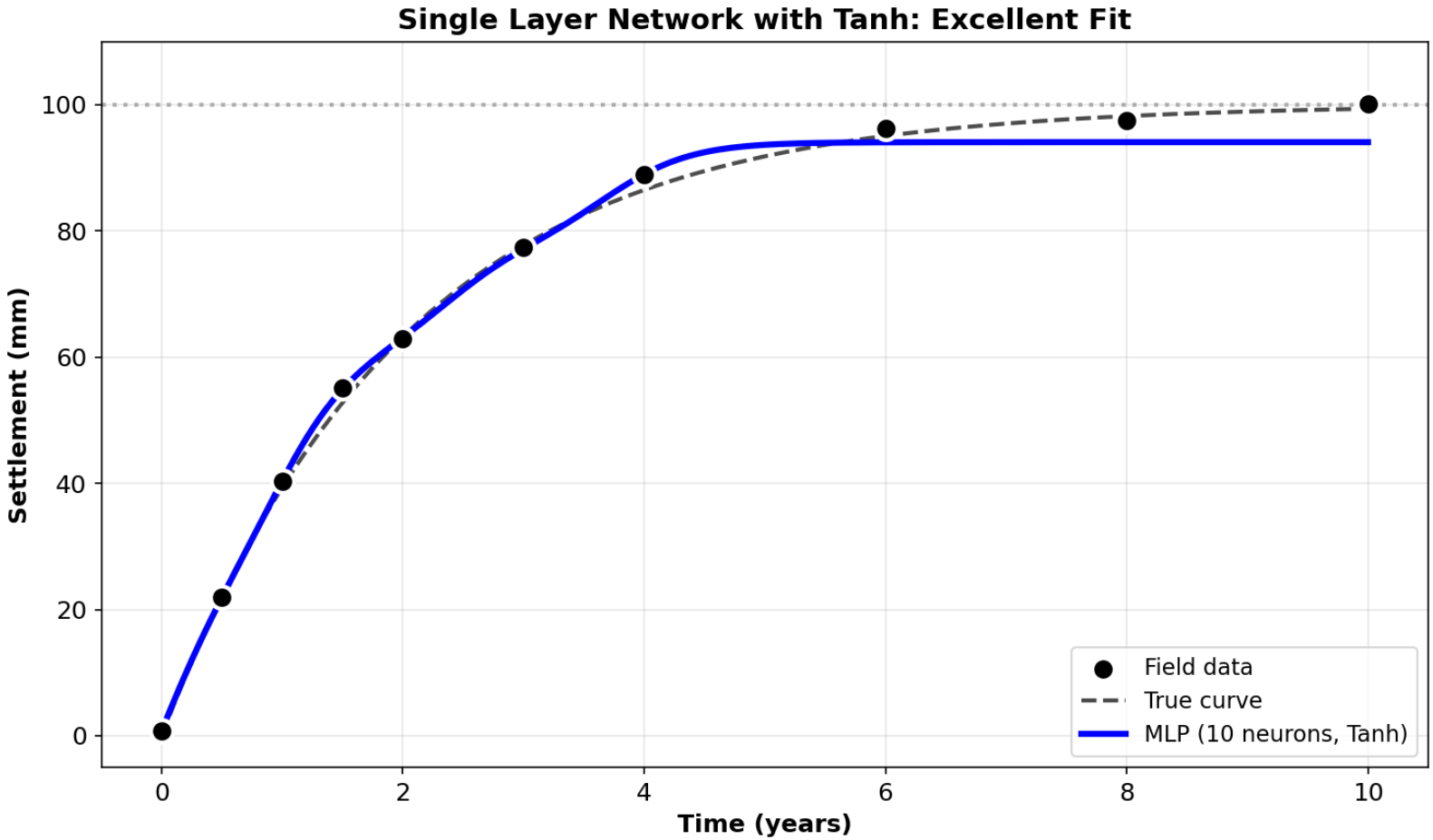
$$\hat{y} = z^{(2)} \quad \text{(final prediction)}$$



Single hidden layer neural network

Single Layer Network on Consolidation

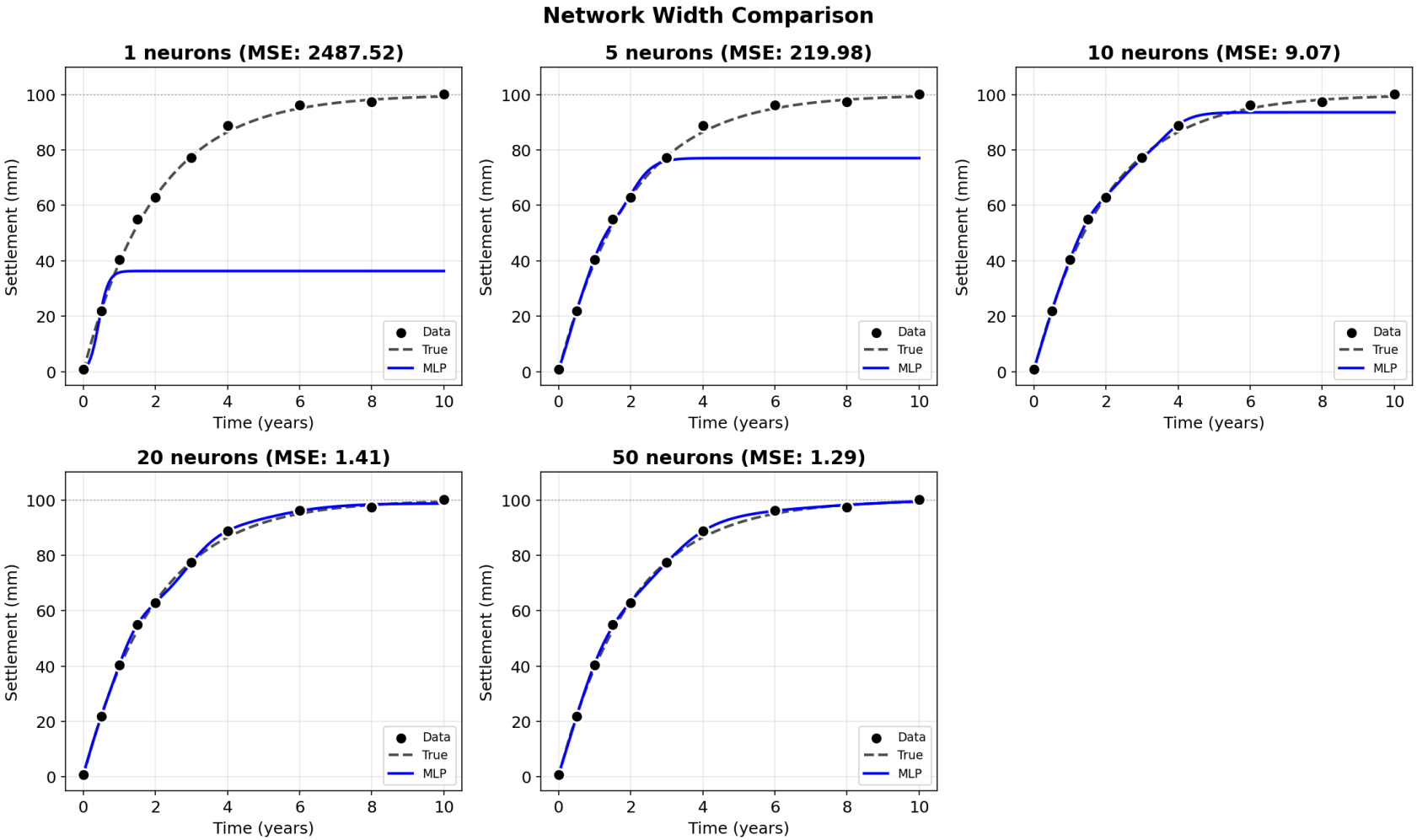
Architecture: Input $\rightarrow$ Hidden (10 neurons, Tanh) $\rightarrow$ Output



Tanh activation fits consolidation curve smoothly

Success: 10 neurons sufficient for smooth approximation of bounded

Network Width Comparison

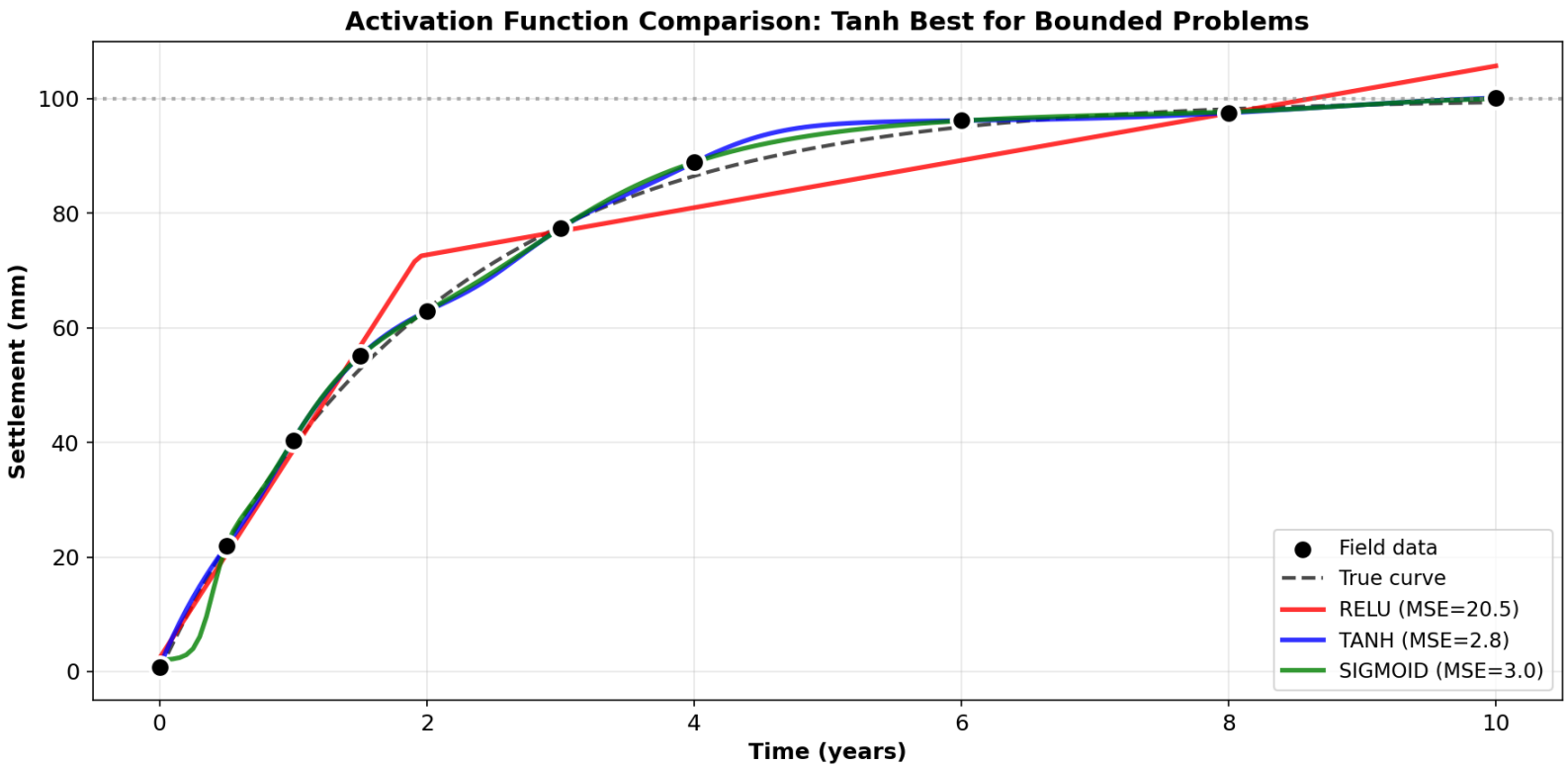


Consolidation approximation: 1, 5, 10, 20, 50 neurons

Finding: 10-20 neurons optimal. Diminishing returns beyond.

Activation Function Choice Matters

Test: ReLU vs Tanh vs Sigmoid on consolidation



Tanh naturally saturates at $S_{\text{final}} = 100$ mm

Rule: Bounded physical problems $\rightarrow$ Tanh/Sigmoid. Unbounded $\rightarrow$ ReLU.

Outline

Training Process

Goal: Find optimal parameters θ^* that minimize loss function.
Loss function (MSE):

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} (u_{NN}(x_i; \theta) - u_i)^2$$

Optimization problem:

$$\theta^* = \arg\min_{\theta} \mathcal{L}(\theta)$$

Training Steps:

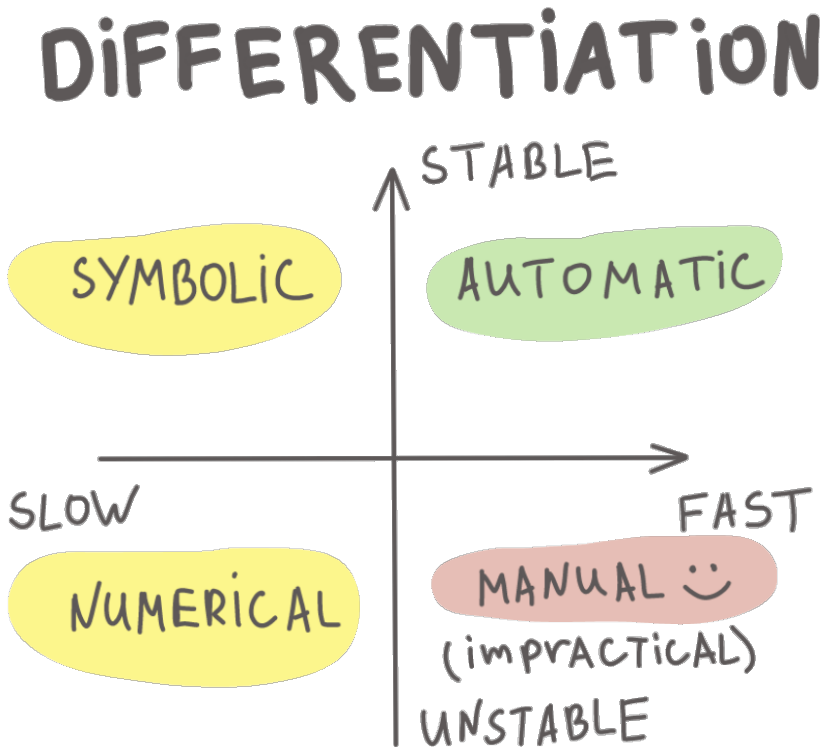
1. Forward pass: compute predictions
2. Calculate loss
3. Backward pass: compute gradients
4. Update parameters
5. Repeat until convergence

The Gradient Problem

Training neural networks requires gradients: $\frac{\partial \mathcal{L}}{\partial \theta}$ for all parameters θ.

Traditional approaches have fundamental flaws:

- **Manual:** Exact, but error-prone and doesn't scale
- **Symbolic:** Exact, but "expression swell"
- **Numerical:** $f'(x) \approx \frac{f(x+h)-f(x-h)}{2h}$
  - Inaccurate (rounding errors)
  - Expensive (multiple evaluations)



For a neural network with thousands of parameters:

$$\theta = \{W^{(1)}, \mathbf{b}^{(1)}, W^{(2)}, \mathbf{b}^{(2)}, \ldots\}$$

We need a fourth approach: Automatic Differentiation!

Automatic Differentiation: The Solution

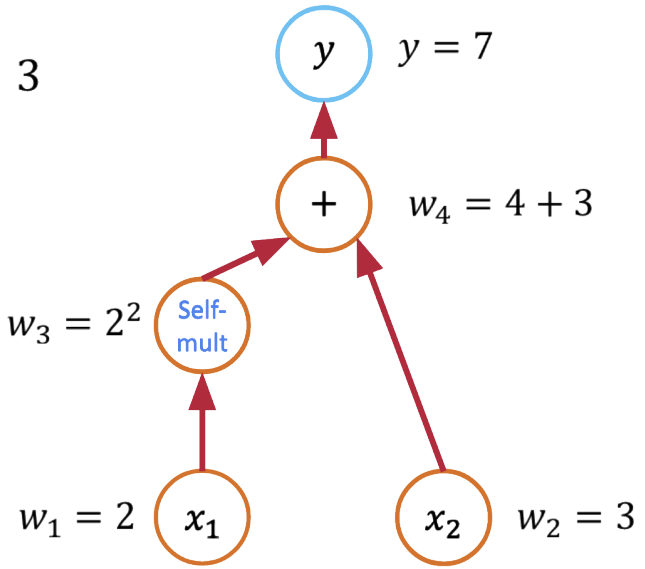
Every function is a computational graph of elementary operations.

Consider: $y = x_1^2 + x_2$

Evaluation Trace:

1. $v_1 = x_1^2$
2. $y = v_1 + x_2$

$x_1 = 2, x_2 = 3$



$y = x_1^2 + x_2$

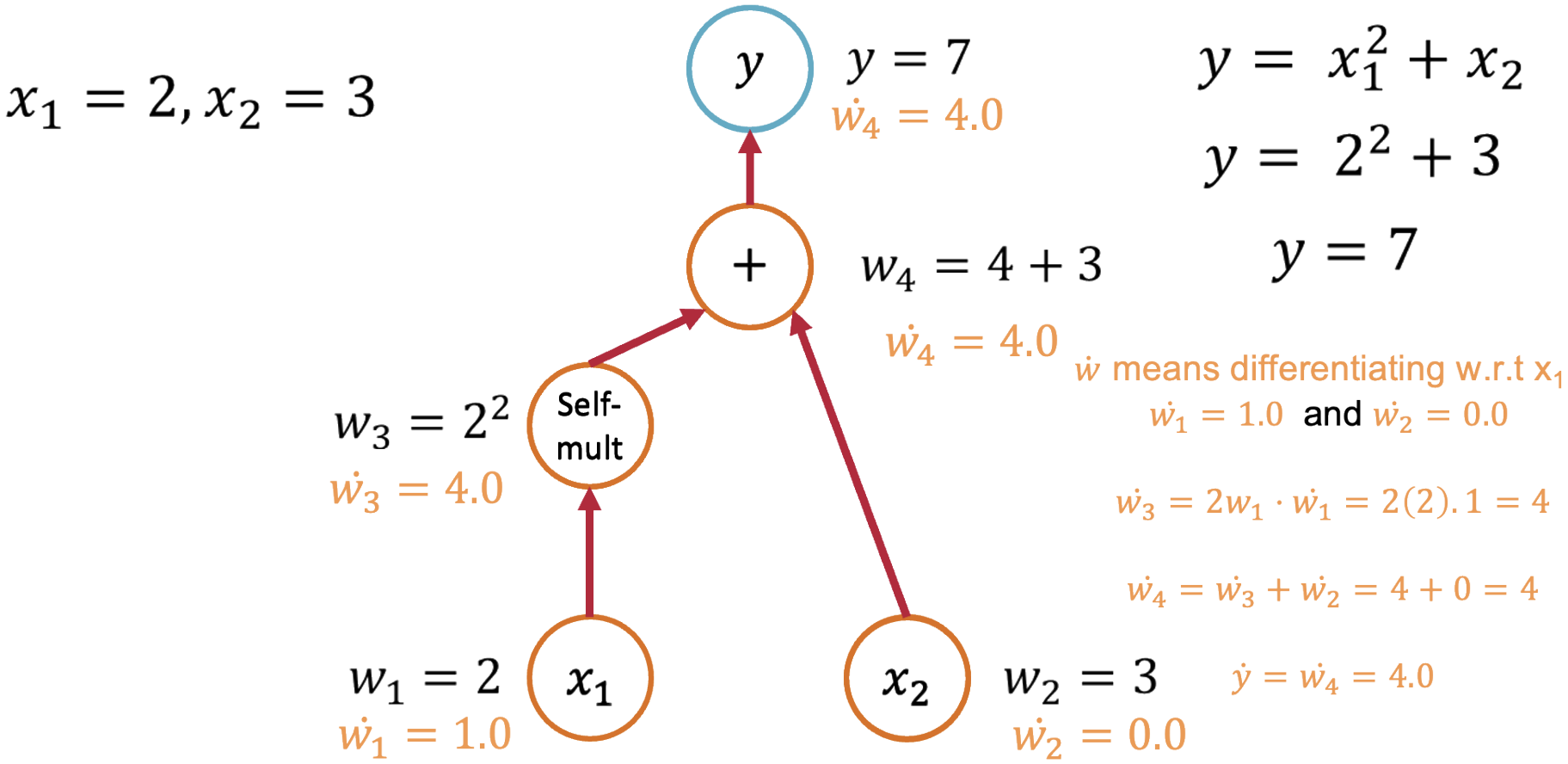
$y = 2^2 + 3$

$y = 7$

This decomposition is the key that makes AD possible. By applying the chain rule to each elementary step, we can compute exact derivatives.

Forward Mode Automatic Differentiation

Forward mode AD propagates derivative information **forward** through the graph, alongside the function evaluation.



▸ AD Demo

Forward Mode: Step-by-Step Example

Let's compute $\frac{\partial y}{\partial x_1}$ for $y = x_1^2 + x_2$.

1. Seed the input w.r.t. x_1

Set $\dot{x}_1 = 1$ and $\dot{x}_2 = 0$.

2. Forward Propagation

- $v_1 = x_1^2 \implies \dot{v}_1 = 2x_1 \cdot \dot{x}_1 = 2x_1 \cdot 1 = 2x_1$
- $y = v_1 + x_2 \implies \dot{y} = \dot{v}_1 + \dot{x}_2 = 2x_1 + 0 = 2x_1$

Result

The final propagated tangent $\dot{y}$ is the derivative: $\frac{\partial y}{\partial x_1} = 2x_1$.

Key Idea: To get the derivative w.r.t. x_2, we would need a *new pass* with seeds $\dot{x}_1 = 0, \dot{x}_2 = 1$.

Reverse Mode Automatic Differentiation

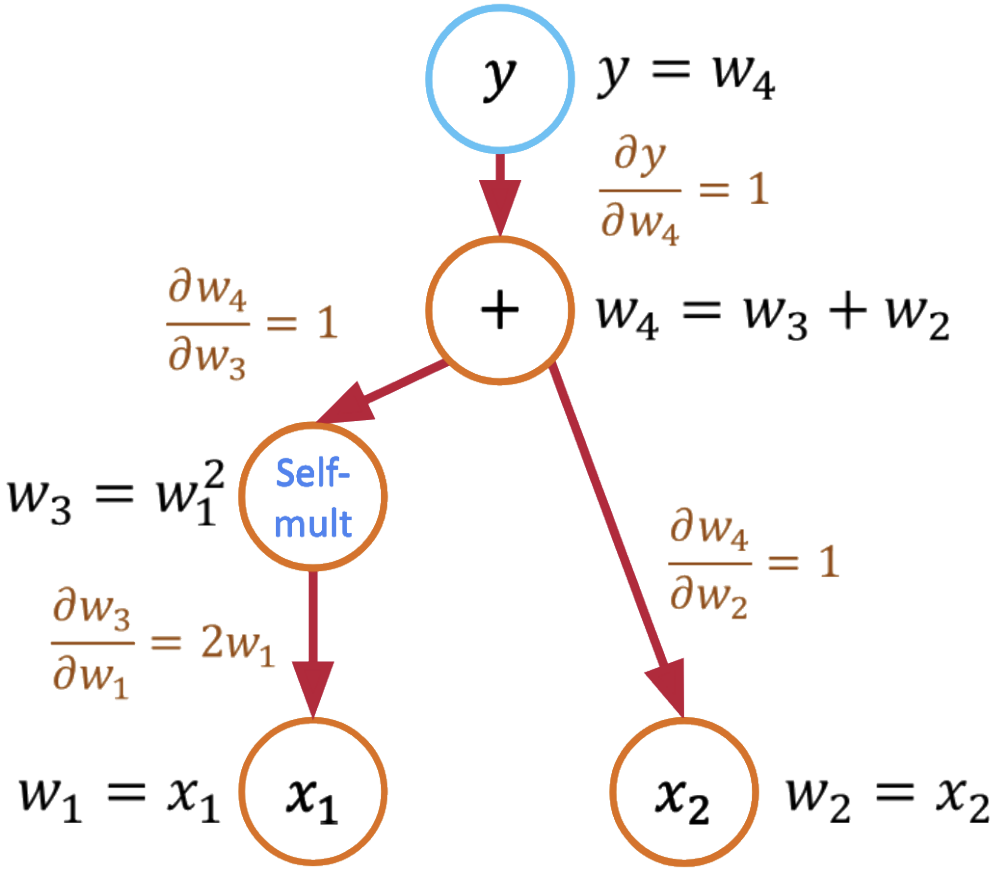
Reverse mode AD (or **backpropagation**) computes derivatives by propagating information **backward** from the output.

Algorithm:

1. **Forward Pass:** Evaluate the function and store all intermediate values.
2. **Backward Pass:** Starting from the output, use the chain rule to propagate "adjoints" ($\bar{v} = \frac{\partial y}{\partial v}$) backward through the graph.

Reverse Mode: One Pass for All Gradients

The backward pass systematically applies the chain rule.
Let's trace the computation for $y = x_1^2 + x_2$:



$$y = x_1^2 + x_2$$

$$\nabla y = \left(\frac{\partial y}{\partial x_1}, \frac{\partial y}{\partial x_2}\right)$$

$$\boxed{\nabla y = (2x_1, 1)}$$

We can find the gradient with respect to the output y

Using $x_1 = 2, x_2 = 3$:

$$\nabla y = (4, 1)$$

Reverse mode computes **all** partial derivatives in a single backward pass.

When to Use Forward vs. Reverse Mode

The choice depends on the dimensions of your function $f : \mathbb{R}^n \to \mathbb{R}^m$.

Forward Mode

- **Cost:** One pass per *input* variable.
- **Efficient when:** Few inputs, many outputs ($n \ll m$).
- **Computes:** Jacobian-vector products ($J \cdot v$).

Reverse Mode

- **Cost:** One pass per *output* variable.
- **Efficient when:** Many inputs, few outputs ($n \gg m$).
- **Computes:** Vector-Jacobian products ($v^T \cdot J$).

Machine Learning context: We have millions of parameters (inputs, n) and a single scalar loss function (output, $m = 1$).

Reverse mode is the clear winner!

▸ AD Notebook

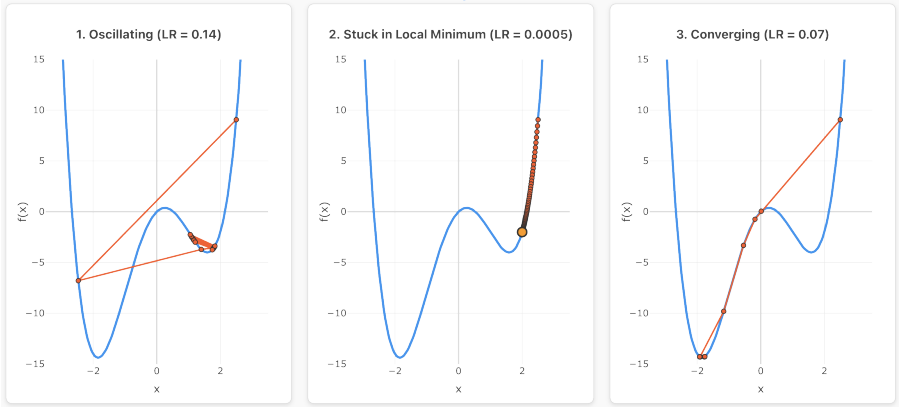
Gradient Descent Algorithm

Basic Algorithm:

1. Initialize weights randomly
2. Loop until convergence:
3. Compute gradient $\frac{\partial \mathcal{L}}{\partial \theta}$
4. Update weights: $\theta \leftarrow \theta - \eta \frac{\partial \mathcal{L}}{\partial \theta}$
5. Return weights

$$\theta_{new} = \theta_{old} - \eta \nabla \mathcal{L}(\theta)$$

where η is the learning rate.



Gradient descent optimization

▸ SGD Demo

Loss Functions

Regression (Consolidation): Mean Squared Error

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(S_i - \hat{S}_i)^2$$

Binary Classification (Liquefaction): Binary Cross-Entropy

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

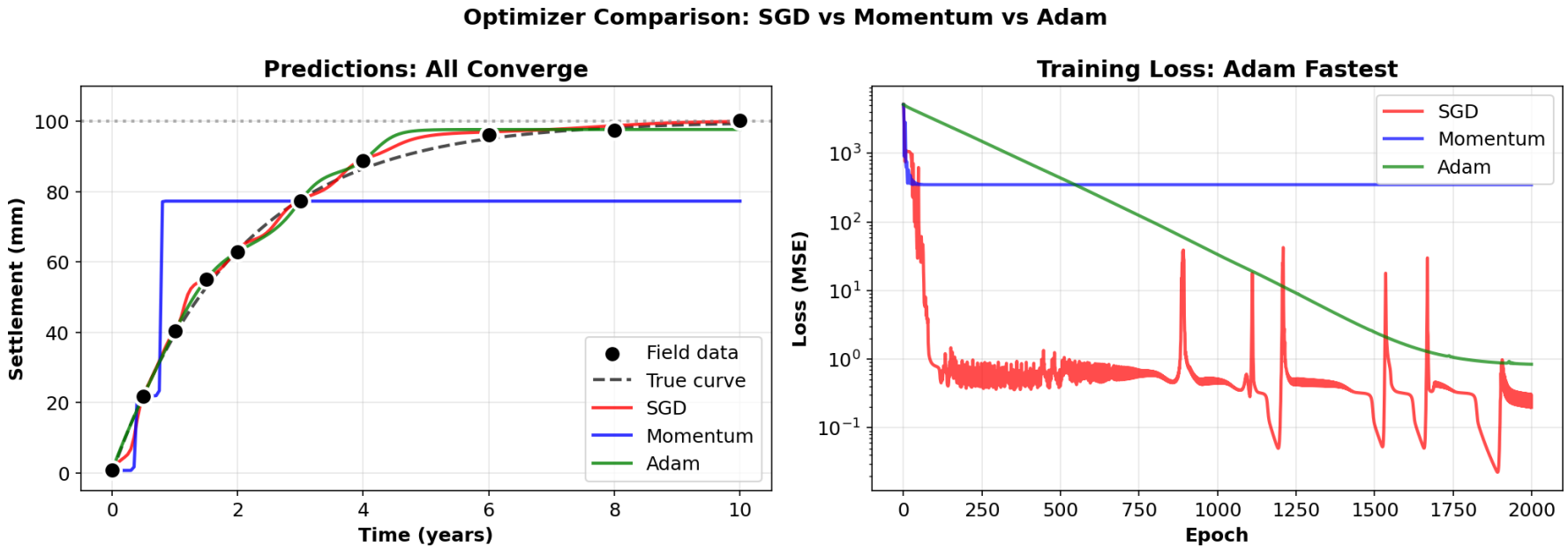
Why BCE for classification?

- Probabilistic interpretation
- Stronger gradients for wrong confident predictions
- Works with sigmoid output $\hat{y} \in (0, 1)$

▸ BCE Demo

Optimizer Comparison

Test on consolidation: SGD vs Momentum vs Adam



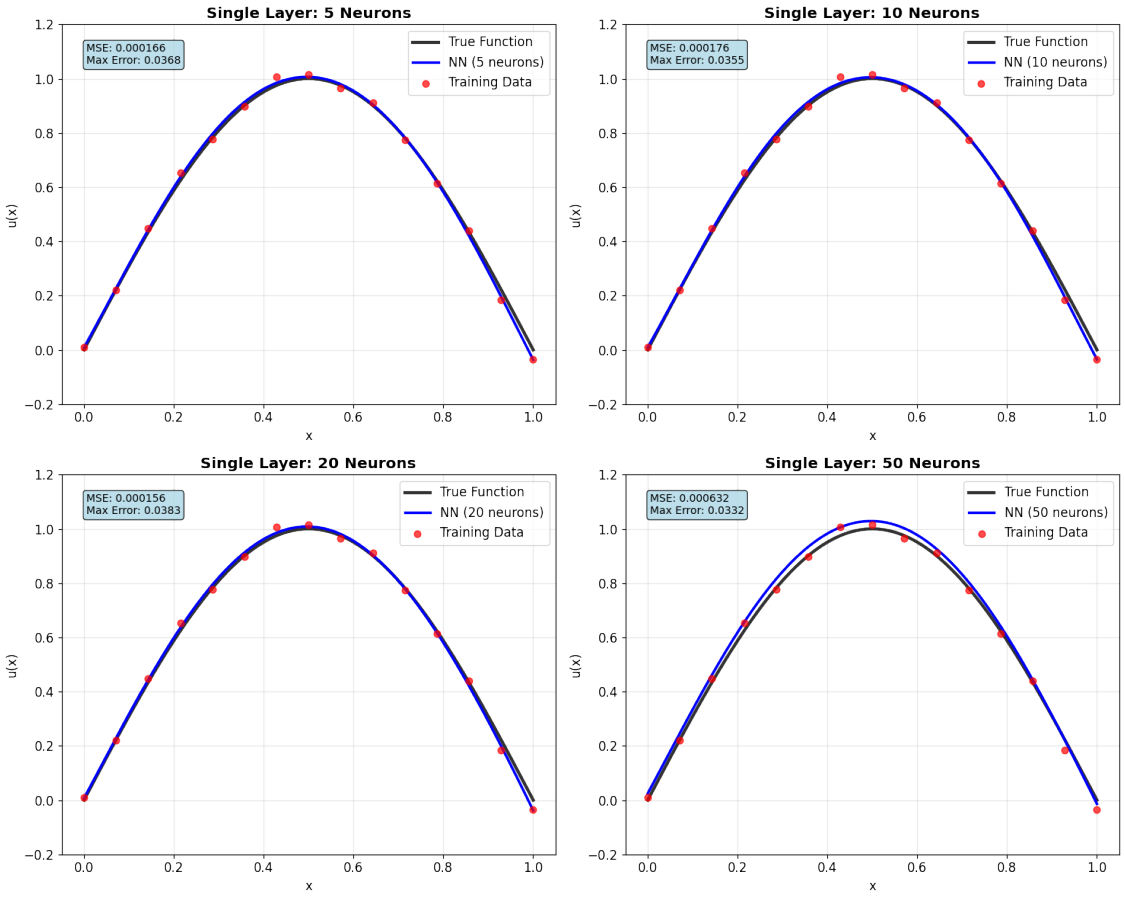
Adam converges fastest, SGD slowest

Recommendation: Adam (lr=0.01) for most problems.

Width vs Approximation Quality

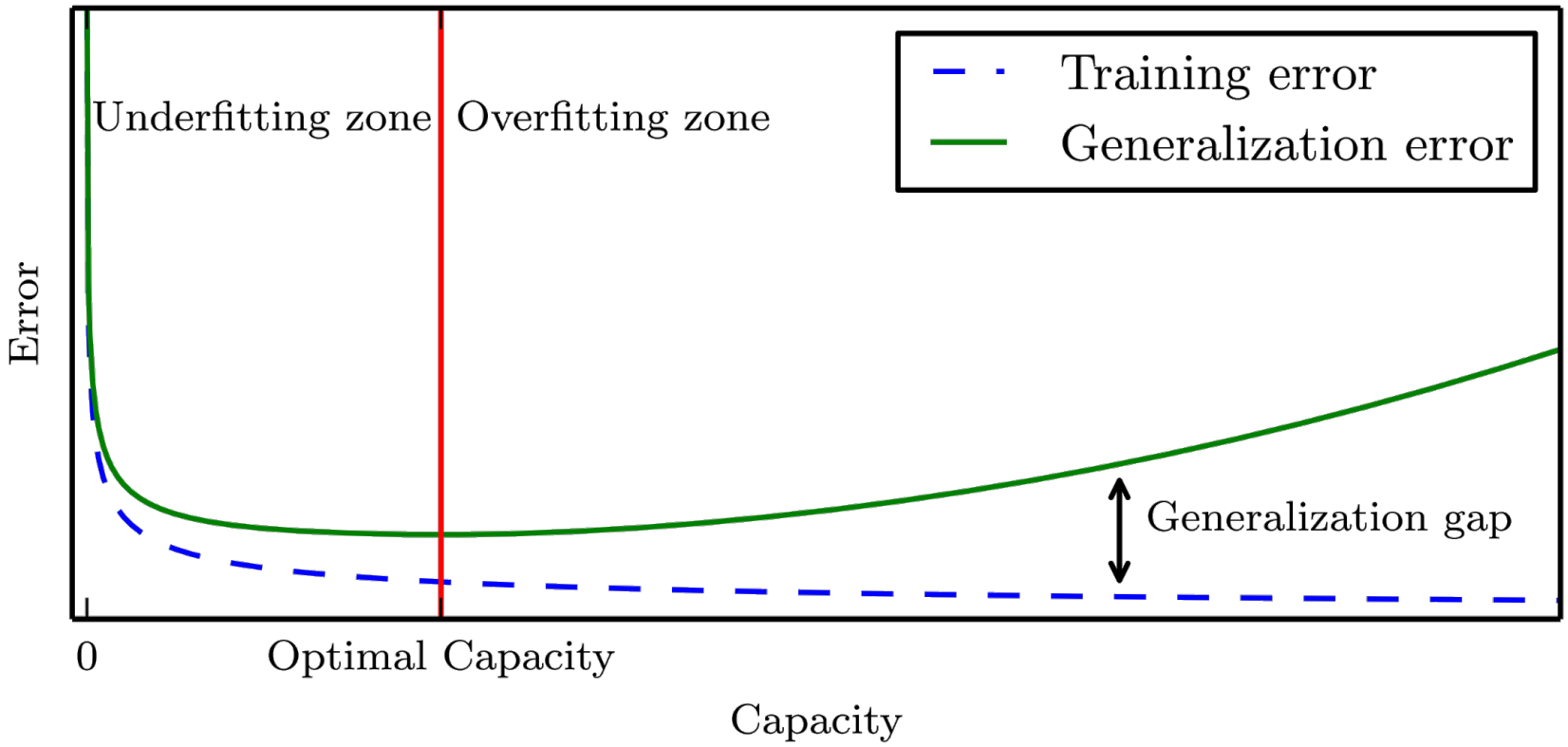
Hypothesis: More neurons $\rightarrow$ better approximation

Experiment: Train networks with 5, 10, 20, 50 neurons



Approximation quality vs network width

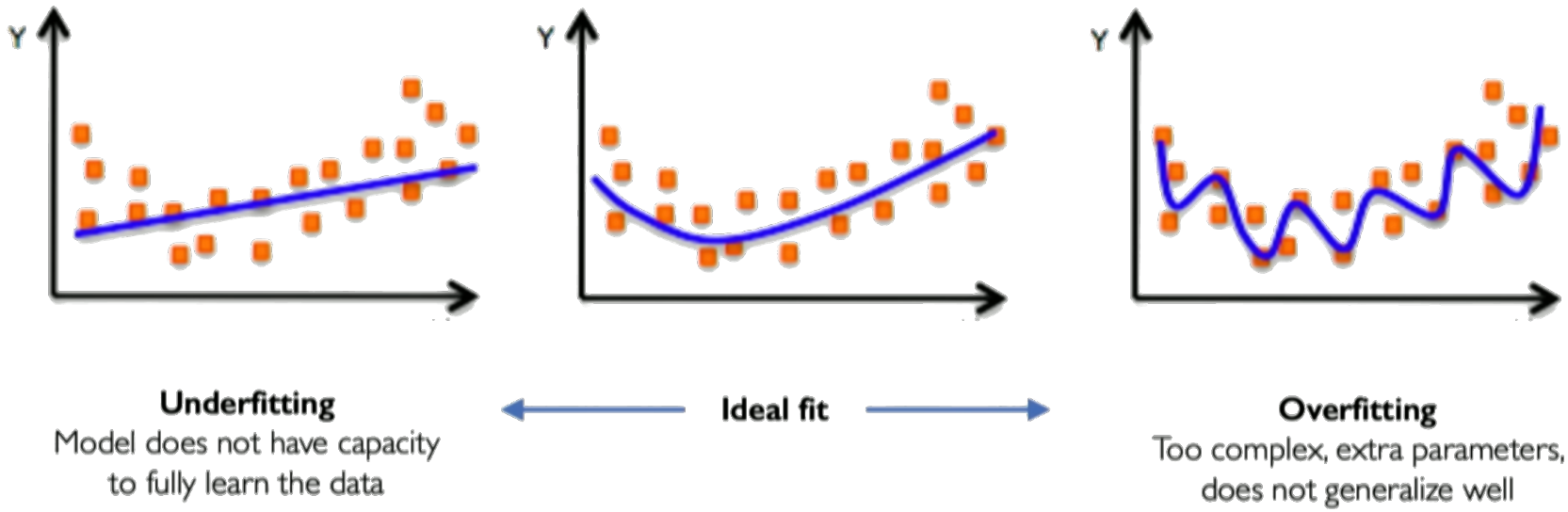
Training/Validation



Training and validation convergence

Overfitting and Model Capacity

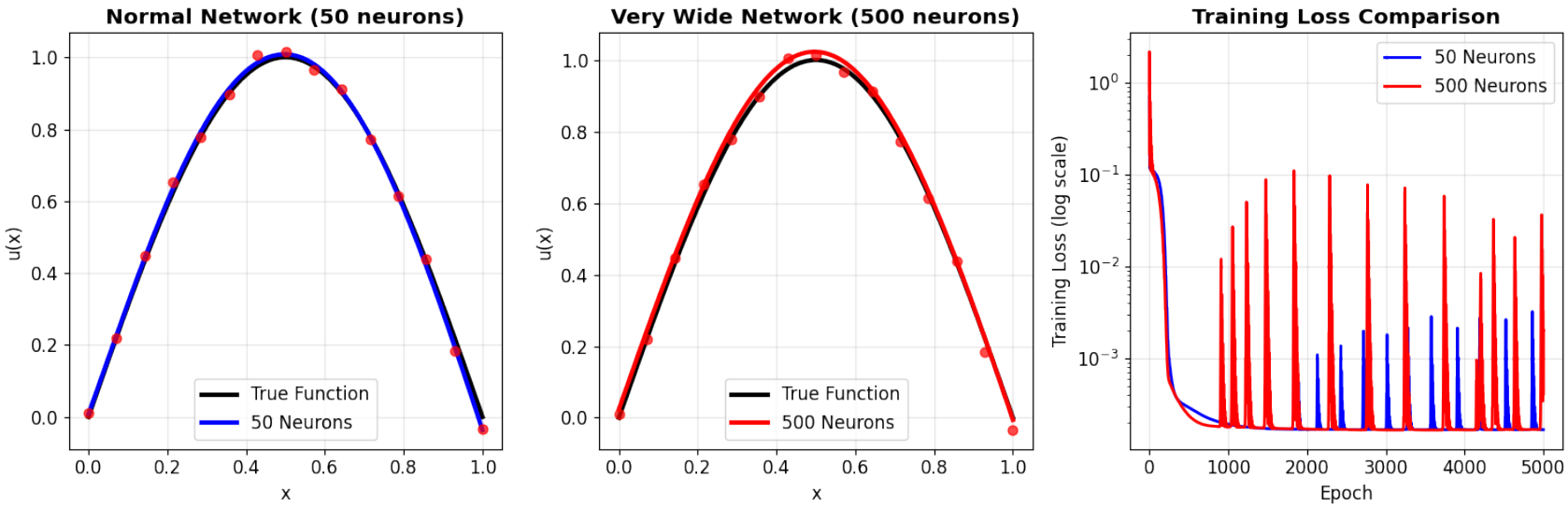
Problem: High-capacity networks can memorize training data.



Under and overfitting illustration

Detection: Monitor validation loss during training.
Solutions: More data, regularization, early stopping, simpler architectures.

Demonstrating Overfitting



Normal (50 neurons) vs very wide (500 neurons) network

Observation: Very wide networks may generalize worse despite lower training loss.

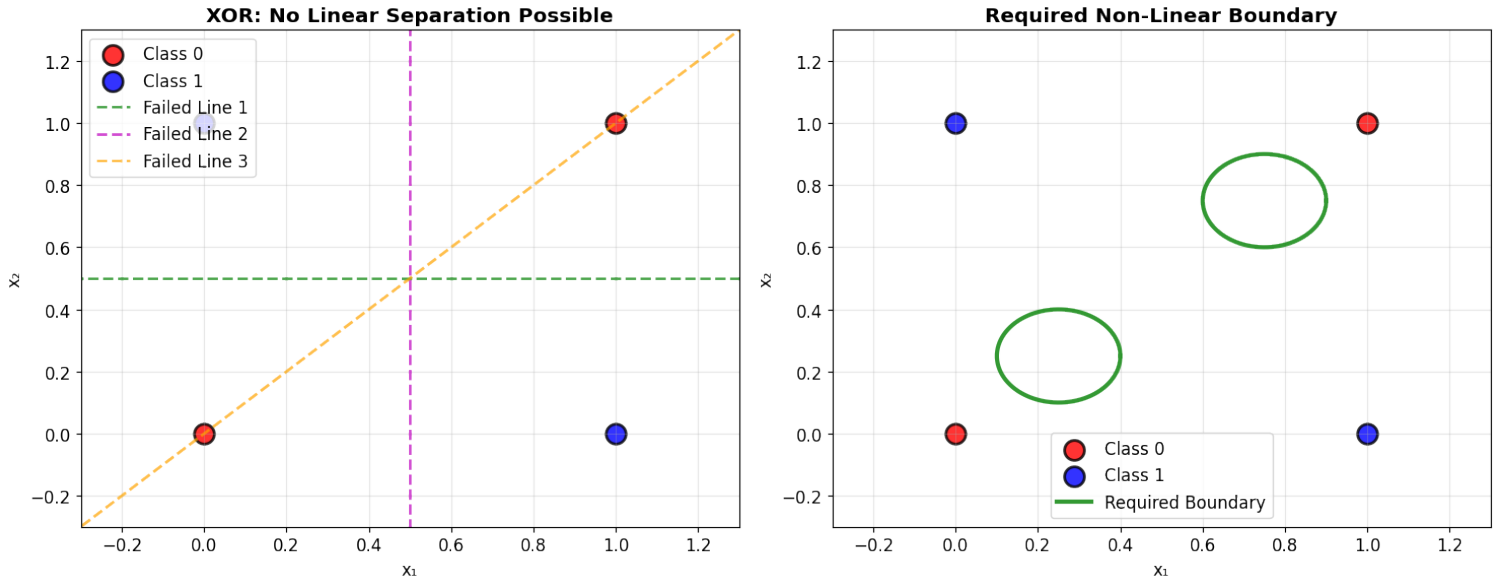
Outline

The XOR Problem: Historical Crisis

XOR Truth Table:

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

The Crisis: No single line can separate these classes!



XOR is not linearly separable

True Single-Layer vs Multi-Layer

Critical Distinction:

- **True Single-Layer:** Input $\rightarrow$ Output (NO hidden layers)
- **Multi-Layer:** Input $\rightarrow$ Hidden $\rightarrow$ Output (1+ hidden layers)

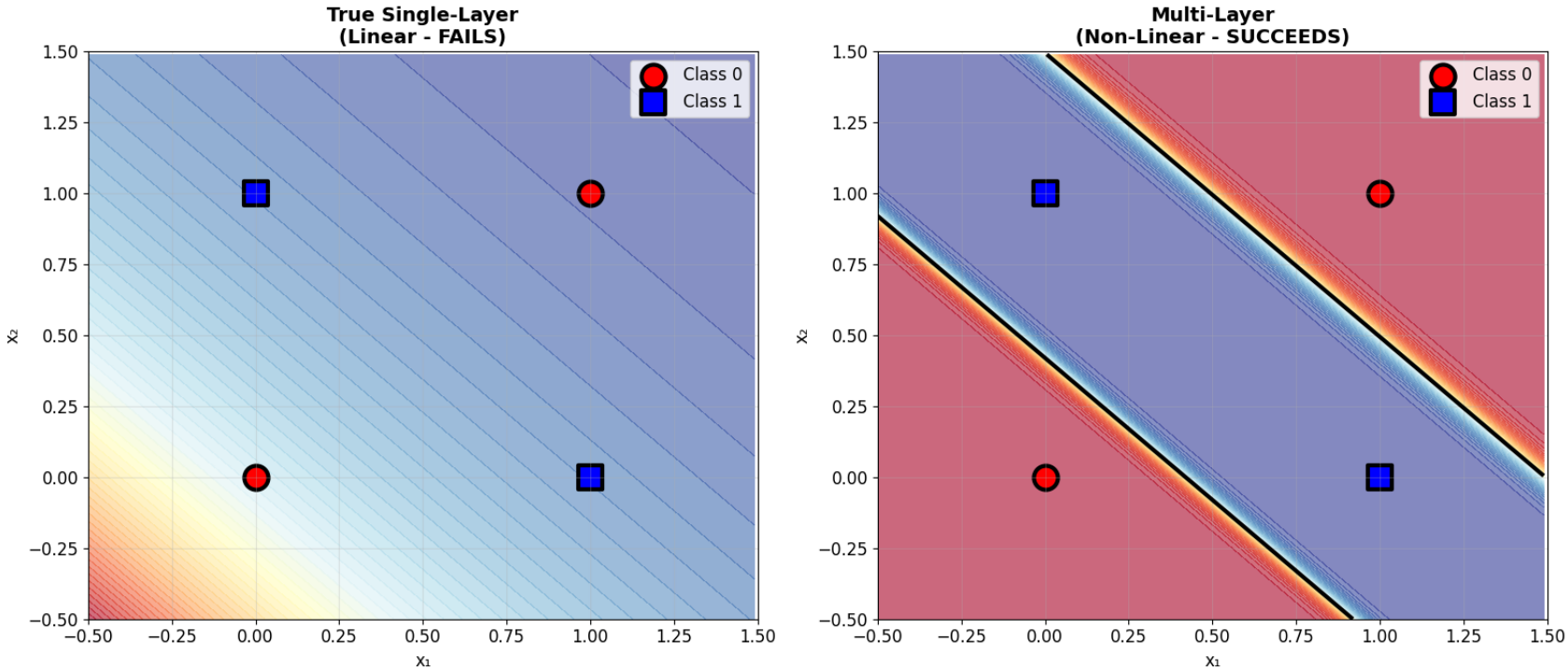
True Single-Layer (fails):

$$y = \sigma(w_1 x_1 + w_2 x_2 + b)$$

Multi-Layer (succeeds):

$$h_1 = \sigma(w_{11} x_1 + w_{12} x_2 + b_1)$$
$$h_2 = \sigma(w_{21} x_1 + w_{22} x_2 + b_2)$$
$$y = \sigma(v_1 h_1 + v_2 h_2 + b_3)$$

XOR Solution: Decision Boundaries



Linear (fails) vs Non-linear (succeeds) decision boundaries

Key Insight: Hidden layers enable curved boundaries through non-linear transformations.

Mathematical Explanation

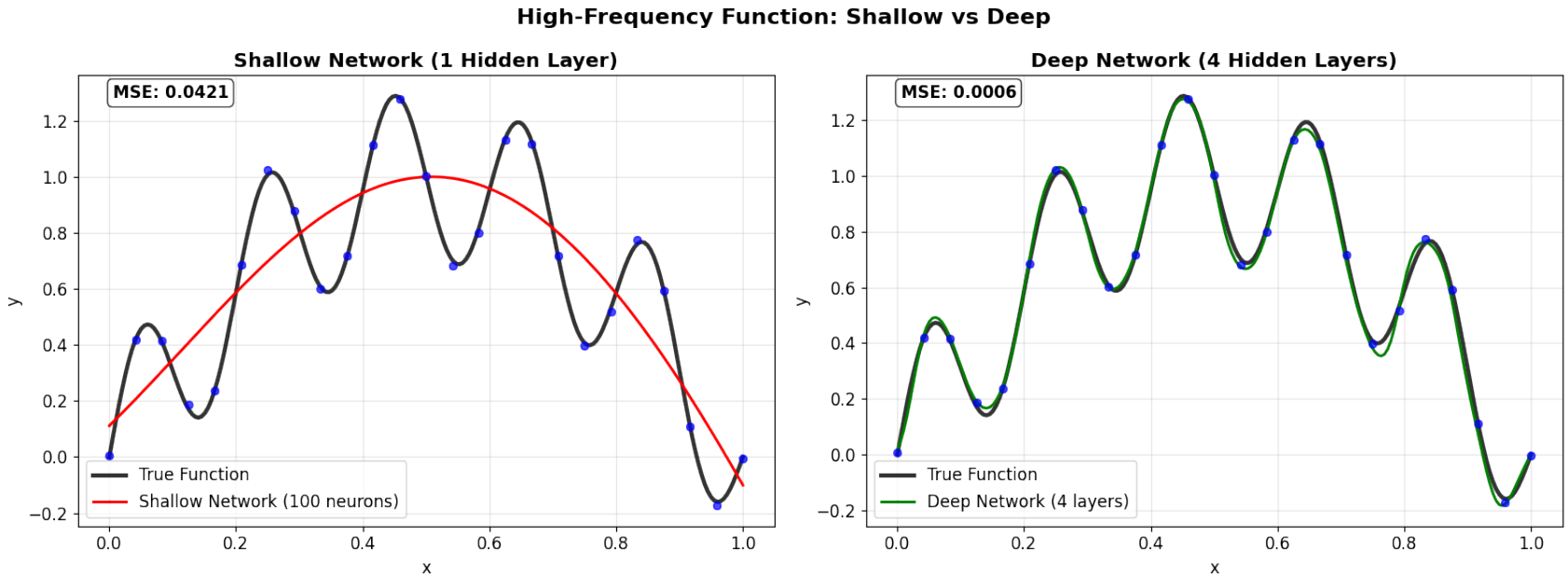
XOR Decomposition:

$$h_1 = \sigma(\text{weights}) \quad (\approx \text{OR gate})$$
$$h_2 = \sigma(\text{weights}) \quad (\approx \text{AND gate})$$
$$y = \sigma(v_1 h_1 + v_2 h_2 + b_3) \quad (\approx \text{OR AND NOT})$$

Result: XOR = (OR) AND (NOT AND) = compositional solution!
General Principle: Complex functions = composition of simple functions.

Beyond XOR: High-Frequency Functions

Test Case: $f(x) = \sin(\pi x) + 0.3\sin(10\pi x)$



Shallow (100 neurons) vs Deep (4 layers, 25 each) networks

Result: Deep networks achieve better performance with fewer parameters.

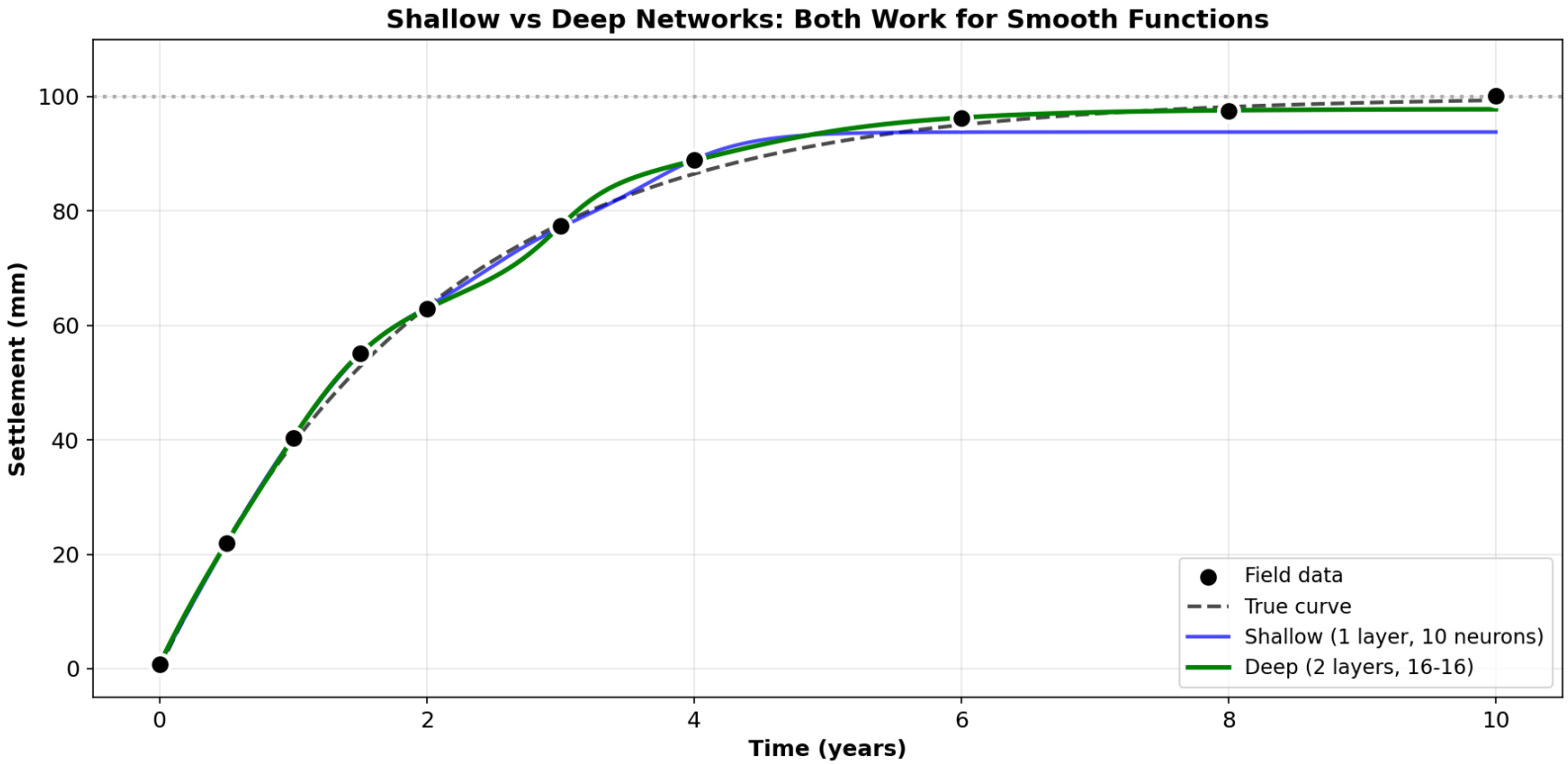
Historical Timeline: From Crisis to Revolution

Year	Event	Impact
1943	McCulloch-Pitts neuron	Foundation laid
1957	Rosenblatt's Perceptron	First learning success
1969	**Minsky & Papert: XOR**	**Showed limits**
1970s-80s	"AI Winter"	Funding dried up
1986	Backpropagation	Enabled multi-layer training
1989	Universal Approximation	Theoretical foundation
2006+	Deep Learning Revolution	Depth proves essential

Lesson: XOR taught us that depth is necessity, not luxury.

Shallow vs Deep on Consolidation

Comparison: 1 layer (10 neurons) vs 2 layers (16-16)



Both work for smooth functions; Deep more parameter-efficient

When to go deep? Complex boundaries, high-frequency, images. Otherwise 1-2 layers sufficient.

Four Key Insights on Depth

1. **Representation Efficiency**
   - Shallow: May need exponentially many neurons
   - Deep: Hierarchical composition is exponentially more efficient
2. **Feature Hierarchy**
   - Layer 1: Simple features (edges, patterns)
   - Layer 2: Feature combinations (corners, textures)
   - Layer 3+: Complex abstractions (objects, concepts)
3. **Geometric Transformation**
   - Each layer performs coordinate transformation
   - Deep networks "unfold" complex data manifolds
4. **Compositional Learning**
   - Complex functions = composition of simple functions
   - Build complexity incrementally

Practical Recipe: 1D Consolidation

Architecture:

- 2 hidden layers, 16-32 neurons each
- Activation: Tanh (bounded), ReLU (unbounded)

Training:

- Optimizer: Adam (lr=0.01)
- Epochs: 2000-3000
- Monitor validation loss for overfitting

Results:

- MSE ¡ 2.0 mm^2
- Smooth predictions over entire domain
- Generalizes well to test data

Summary and Conclusions

Key Takeaways:

- Nonlinearity essential for complex patterns
- UAT: Single layer theoretically sufficient
- Practice: 1-2 layers for most problems
- Activation choice matters: Tanh for bounded, ReLU for unbounded
- Training: Adam + validation prevents overfitting
- XOR lesson: Depth creates feature hierarchies

Applications:

- Consolidation settlement prediction
- Liquefaction classification
- Function approximation from sparse data
- Physics-informed neural networks (PINNs)

Questions?

Thank you!

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